

JILL WANG

Data Scientist & AI/ML Engineer

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Kaggle: kaggle.com/jillwang87 · **Portfolio:** jillwang87.github.io

WORK EXPERIENCE

Head of Analytics

Jul 2023 – Present

Truthifi – digital finance monitoring platform

- Built and shipped 20+ MCP tools serving an internal AI agent and external third-party MCP clients, powering a complete, customizable financial health analysis experience
- Engineered prompt safety guardrails for the internal agent, rich metadata contracts for third-party MCP clients, user friendly tool inputs, and MongoDB-backed pagination to optimize query load for large result sets
- Shipped directive MCP tools and tier-based upgrade nudges that drove account-linking conversion, subscription growth, and rising assets under monitoring
- Built PostHog dashboards and SQL-driven funnel analyses across the public site and core app to pinpoint signup and account-linking drop-off and drive fixes
- Built efficient ETL pipelines for vendor APIs (e.g. Xignite, Morningstar, Yahoo Finance) and government sources (e.g. SEC, MSRB) that meticulously standardize, validate, and enhance millions of records to maintain the quality of our market data collections
- Developed real-time performance feedback and diagnostic metrics to benchmark user's investment portfolios
- *Technology: TypeScript, MongoDB, MCP, React, Node.js, NestJS, GraphQL*

Data Scientist (via Adastra)

Mar 2023 – Jun 2023

Mark Anthony Group – alcoholic beverage company

- Built an XGBoost sales-forecasting ML model on PySpark pipelines processing millions of rows to power gap and attention marketing analysis
- *Technology: Python, AWS SageMaker, XGBRegressor, SQL*

Operation Research Analyst (via Adastra)

Aug 2022 – Feb 2023

Taiga Building Products Ltd. – wholesale building materials distributor

- Built an address-clustering AI model (Google Direction API) and a truck-load optimization model by mathematical formulation (Google OR-Tools) to optimize delivery routing and truck-loads — reducing freight costs and improving sales margins
- *Technology: Python, Azure ML, Google OR-Tools, Google Direction API*

Data Analytics Developer (via Adastra)

May 2021 – Jun 2022

TD Bank – project delivery team

- Built a Python NLP/ML pipeline (data lifecycle management application) using computational linguistics and NLP to extract features that predict security classifications and retention periods from company digital assets — delivering predictions in minutes that previously took months of manual work from data stewards
- Built ETL pipelines on the Apache big data ecosystem feeding real-time data straight from source systems into 50+ interactive Tableau executive dashboards, engineering custom formulas to handle the underlying calculations
- *Technology: Impala, Python, scikit-learn, NLP, spaCy, tkinter, Tableau, Oozie*

Data Science Consultant

Apr 2021 – Jun 2023

Adastra North America – data and AI consultancy

- Delivered scalable analytics and machine learning solutions across enterprise engagements at TD Bank, Taiga Building Products, and Mark Anthony Group

Data Scientist & Software Developer (Intern)

Jun 2019 – Aug 2019

Graphen – R&D group developing an anti-money-laundering platform

- Developed an unsupervised model on time-series data and a supervised model on multivariate data for financial behaviour anomaly/fraud detection, applying statistical methods to design a systematic risk (anomaly) score and a numerical analysis method (power spectral density) to improve time series model accuracy
- Developed the front end of a React-based anti-money-laundering platform
- *Technology: Python, NumPy, pandas, scikit-learn, XGBoost, JavaScript, React*

SKILLS

AI Engineering: MCP · LangChain · LangSmith · OpenAI API

Data Science: Python · NumPy & pandas · scikit-learn · XGBoost · TensorFlow/Keras · spaCy · statsmodels · SciPy · R · MATLAB

ETL: MongoDB · SQL · Hadoop · Impala · Hive · Oozie

Business Intelligence: Tableau · PostHog

Software Development: TypeScript · React · JavaScript · Node.js · NestJS · GraphQL

Others: Git · Bash · C/C++

EDUCATION

Honours Bachelor of Mathematics in Computational Mathematics

University of Waterloo, 2021

Honours Bachelor of Mathematics in Statistics, Computing Technology Option

University of Waterloo, 2021

Relevant Coursework: Statistical Learning & Classification, Generalized Linear Models & Applications, Applied Probability, Computational Statistics and Data Analysis, Mathematical Statistics, Data Structures & Algorithms, Object-Oriented Software Development, Applied Cryptography, Computational Methods for Differential Equations